

Principles of Programming Languages:

Problem Sheet 4 - Continuations and types

Wira Azmoon Ahmad

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Question 1

```
1 type Answer = [Int]
2
3 revk :: [Int] -> ([Int] -> Answer) -> Answer
4 revk [] k = k []
5 revk (x:xs) k = revk xs (\ys -> k (ys ++ [x]))
6
7 revit :: [Int] -> [Int] -> [Int]
8 revit [] ys = ys
9 revit (x:xs) ys = revit xs ([x] ++ ys)
```

Suppose we have a list $[a_1, a_2, \dots, a_n]$. We show by induction that the continuation for iteration i can be written in the form

$$(\dots((y_i ++ [a_{i-1}]) ++ [a_{i-2}]) ++ \dots ++ [a_1]) = y_i + [a_{i-1}, \dots, a_1]$$

```
1 -- Q1
2 -- Base case: n = 0
3 revk [] id = id [] = []
```

Suppose for $n = k$, the continuation is

$$y_k + [a_{k-1}, \dots, a_1]$$

```
1 -- n = k
2 revk [a_k, a_{k+1}, ..., a_n] (\yk -> yk ++ [a_{k-1}, ..., a_1])
3 = revk [a_{k+1}, ..., a_n] (\yk1 -> (\yk -> yk + [a_{k-1}, ..., a_1]) (yk1 ++ a_k))
4 = revk [a_{k+1}, ..., a_n] (\yk1 -> (yk1 ++ [a_k]) ++ [a_{k-1}, ..., a_1])
5 = revk [a_{k+1}, ..., a_n] (\yk1 -> yk1 ++ [a_k, a_{k-1}, ..., a_1])
```

We write in fun:

```
1 val revk(xss) =
2   let val yss = new() in let val ys = new() in
3     yss := xss; ys := nil;
4     while !yss <> nil do
5       (ys := head(!yss) : !ys ; yss := tail(!yss));
6     !ys;;
```

Question 2

```
1 type Answer = Int
2 data Cont = Show | Add Int Cont | FibAdd Int Cont
3
4 appcont :: Cont -> Int -> Answer
5 appcont Show r = r
6 appcont (Add n k) r = appcont k (r + n)
7 appcont (FibAdd n k) r = fibk (n-2) (Add r k)
8
9 fibk :: Int -> Cont -> Answer
10 fibk n k = if n <= 1 then appcont k n else fibk (n-1) (FibAdd n k)
```

```
1
2 -- Q2
3 fibk 4 Show
4 = fibk 3 (FibAdd 4 Show)
5 = fibk 2 (FibAdd 3 (FibAdd 4 Show))
6 = fibk 1 (FibAdd 2 (FibAdd 3 (FibAdd 4 Show)))
7 = appcont (FibAdd 2 (FibAdd 3 (FibAdd 4 Show))) 1
8 = fibk 0 (Add 1 (FibAdd 3 (FibAdd 4 Show)))
9 = appcont (Add 1 (FibAdd 3 (FibAdd 4 Show))) 0
10 = appcont (FibAdd 3 (FibAdd 4 Show)) 1
11 = fibk 1 (Add 1 (FibAdd 4 Show))
12 = appcont (Add 1 (FibAdd 4 Show)) 1
13 = appcont (FibAdd 4 Show) 2
14 = fibk 2 (Add 2 Show)
15 = fibk 1 (FibAdd 2 (Add 2 Show))
16 = appcont (FibAdd 2 (Add 2 Show)) 1
17 = fibk 0 (Add 1 (Add 2 Show))
18 = appcont (Add 1 (Add 2 Show)) 0
19 = appcont (Add 2 Show) 1
20 = appcont Show 3
21 = 3
22
23 fib 4
24 = fib 3 + fib 2
25 = (fib 2 + fib 1) + fib 2
26 = ((fib 1 + fib 0) + fib 1) + fib 2
27 = ((1 + 0) + fib 1) + fib 2
28 = (1 + fib 1) + fib 2
29 = (1 + 1) + fib 2
30 = 2 + fib 2
31 = 2 + (fib 1 + fib 0)
32 = 2 + (1 + 0)
```

Question 3

Charlie is right because the point of CPS is to prevent unbounded recursion, and the division here has no recursion.

Question 5

On top of the transition rules in lecture, we have the following

$$\begin{aligned}
 \langle succ, env, k \rangle &\rightarrow [k, succ] \\
 \langle pred, env, k \rangle &\rightarrow [k, pred] \\
 [succ(\square) : k, v] &\rightarrow \langle v + 1, -, k \rangle \\
 [pred(\square) : k, v] &\rightarrow \langle v - 1, -, k \rangle
 \end{aligned} \tag{1}$$

$$\begin{aligned}
 &\langle pred(succ(1)), -, Show \rangle \\
 &\rightarrow \langle pred, -, (\square(succ(1)), -) : Show \rangle \\
 &\rightarrow [(\square(succ(1)), -) : Show, pred] \\
 &\rightarrow \langle succ(1), -, pred(\square) : Show \rangle \\
 &\rightarrow \langle succ, -, (\square(1), -) : pred(\square) : Show \rangle \\
 &\rightarrow [(\square(1), -) : pred(\square) : Show, succ] \\
 &\rightarrow \langle 1, -, succ(\square) : pred(\square) : Show \rangle \\
 &\rightarrow [succ(\square) : pred(\square) : Show, 1] \\
 &\rightarrow \langle 2, -, pred(\square) : Show \rangle \\
 &\rightarrow [pred(\square) : Show, 2] \\
 &\rightarrow \langle 1, -, Show \rangle
 \end{aligned} \tag{2}$$

Question 6

(a) No to both. Consider the infinite chain

$$1 \leq 2 \leq 3 \leq \dots$$

which clearly does not have any upper bound. It has no bottom element since for any integer x , we can also find an integer $y = x - 1$ so $x \not\leq y$.

(b) Take the empty set $\emptyset \subset \mathbb{Z}$. It is vacuously true that $\forall y \in \mathbb{Z}, \forall x \in \emptyset, y \leq x$, so all $y \in \mathbb{Z}$ are upper bounds for any chain from $\emptyset$, and since $\mathbb{Z}$ has no least element, there is no least upper bound.

(c)

Proposition 3. *For any chain $S_1 \subseteq S_2 \subseteq \dots \subseteq S_n \subseteq \dots$, the set $S = \bigcup_i S_i$ is the least upper bound.*

Proof. Clearly, $\forall i, S_i \subseteq S$. So S is an upper bound.

Suppose we have another upper bound S_y . As it is an upper bound, $\forall i, S_i \subseteq S_y$. Then $\bigcup_i S_i \subseteq S_y$. By definition, $S = \bigcup_i S_i \subseteq S_y$. So S is the least upper bound. $\square$

The bottom element is $\perp = \emptyset$ since $\forall S_i \in P(\mathbb{N}), \emptyset \subseteq S_i$.

Proposition 4. *$F(S) = \{1\} \cup \{n + 2 | n \in S\}$ is continuous.*

Proof. Suppose $S_x \subseteq S_{x'}$. Then $F(S_x) = \{1\} \cup \{n + 2 | n \in S_x\} \subseteq \{1\} \cup \{n + 2 | n \in S_{x'}\} = F(S_{x'})$. So F is monotone.

Suppose we have a chain $S_1 \subseteq S_2 \subseteq \dots$. We show that $F(\bigcup_i S_i) = \bigcup_i F(S_i)$.

$$\begin{aligned} F\left(\bigcup_i S_i\right) &= \{1\} \cup \{n + 2 | n \in \bigcup_i S_i\} \\ &= \{1\} \cup \bigcup_i \{n + 2 | n \in S_i\} \\ &= \bigcup_i \{1\} \cup \{n + 2 | n \in S_i\} \\ &= \bigcup_i F(S_i) \end{aligned} \tag{3}$$

So F preserves least upper bounds of chains.

Then F is continuous. $\square$

By Tarski's fixed point theorem, the least fixed point of F is

$$\bigcup_n F^n(\emptyset) = \{k | 2k - 1 \in \mathbb{N}\}$$

In other words, it is set of odd natural numbers.